The University of Texas at Austin
In-Class Work 18

M339D Introduction to Financial Mathematics

November 14, 2026

Instructions: Provide your complete solution to the following problems. Final answers only, without appropriate justification, will receive zero points even if correct.

Log-Normal Stock Prices: Tail Probabilities.

Motivation.

Problem 18.1: *Let the stock prices be modeled using the lognormal distribution. Under the risk-neutral probability measure, the mean stock price at time*-1 *equals* 120 *and the median stock price* 115*. What is the risk-neutral probability that the time*-1 *stock price exceeds* 100*?*

Solution: *The stock price at time*-1 *is lognormally distributed. In fact, using our usual parameters, we can rewrite it as*

$$S(1) = S(0)e^{(r-\frac{1}{2}\sigma^2)+\sigma Z}.$$

Recall that the median of $S(1)$ *equals* $S(0)e^{(r-\frac{1}{2}\sigma^2)}$*. So, the required probability can be expressed as*

$$\begin{aligned}\mathbb{P}[S(1) > 100] = \mathbb{P}[115e^{\sigma Z} > 100] &= \mathbb{P}\left[Z > \frac{1}{\sigma}\ln\left(\frac{100}{115}\right)\right] \\ &= \mathbb{P}\left[Z < \frac{1}{\sigma}\ln\left(\frac{115}{100}\right)\right] = N\left(\frac{1}{\sigma}\ln\left(\frac{115}{100}\right)\right).\end{aligned}$$

Since the mean of $S(1)$ *equals* $S(0)e^{(\alpha-\delta)}$*, we have*

$$e^{\frac{1}{2}\sigma^2} = \frac{120}{115} \quad \Rightarrow \quad \sigma = \sqrt{2\ln(1.04348)} = 0.2918.$$

So, our final answer is

$$\mathbb{P}[S(1) > 100] = N(0.48) = 0.6844.$$

Derivation.

Problem 18.2: *Consider a non-dividend-paying stock. What is the probability that the stock outperforms a risk-free investment* under the risk-neutral probability measure $\mathbb{P}^*$*?*

Solution: *Let the initially invested amount, either way, be* $S(0)$*.*

- *If it's a risk-free investment, then the balance at time-*T *equals* $S(0)e^{rT}$*.*

- *If it's a stock investment. We have sufficient funds to purchase one share at time-*0*. We still own that share at time-*T *and its market price is* $S(T)$*. So, that's our wealth.*

We are tasked with figuring out the following probability

$$\mathbb{P}^*[S(T) > S(0)e^{rT}].$$

Remark 18.3*: Note that this question is equivalent to the one of whether the stock investment is positive under the risk-neutral probability measure* $\mathbb{P}^*$*.*

In the Black-Scholes model, we can express the time-T *stock price as*

$$S(T) = S(0)e^{\left(r-\frac{\sigma^2}{2}\right)T+\sigma\sqrt{T}Z} \quad \text{with} \quad Z \sim N(0,1).$$

Substituting the Black-Scholes $S(T)$ *into the probability we seek, we get*

$$\mathbb{P}^*\left[S(0)e^{\left(r-\frac{\sigma^2}{2}\right)T+\sigma\sqrt{T}Z} > S(0)e^{rT}\right] = \mathbb{P}^*\left[e^{\left(r-\frac{\sigma^2}{2}\right)T+\sigma\sqrt{T}Z} > e^{rT}\right].$$

Since the natural logarithm is increasing, the above probability equals

$$\begin{aligned}\mathbb{P}^*\left[\left(r-\frac{\sigma^2}{2}\right)T+\sigma\sqrt{T}Z > rT\right] &= \mathbb{P}^*\left[-\frac{\sigma^2}{2}T+\sigma\sqrt{T}Z > 0\right] \\ &= \mathbb{P}^*\left[\sigma\sqrt{T}Z > \frac{\sigma^2}{2}T\right] = \mathbb{P}^*\left[Z > \frac{\sigma\sqrt{T}}{2}\right].\end{aligned}$$

Due to the symmetry of $N(0,1)$ *with respect to the vertical axis, the above probability equals* $N\left(-\sigma\frac{\sqrt{T}}{2}\right)$ *where* N *denotes the standard normal cumulative distribution function.*

Remark 18.4*: As* $T \to \infty$*, the above probability converges to* 0*.*

Problem 18.5: *Consider a non-dividend-paying stock and European call option on that stock. As usual, let* K *denote the option's strike price and let* T *denote its exercise date.* Under the risk-neutral probability measure $\mathbb{P}^*$*, what is the probability that the call is* ***in-the money*** *at time-*T*?*

Solution: *This derivation retraces the steps of the previous one. With* $Z \sim N(0,1)$*, we have*

$$\begin{aligned}\mathbb{P}^*[S(T) > K] &= \mathbb{P}^*\left[S(0)e^{\left(r-\frac{\sigma^2}{2}\right)T+\sigma\sqrt{T}Z} > K\right] \\ &= \mathbb{P}^*\left[e^{\left(r-\frac{\sigma^2}{2}\right)T+\sigma\sqrt{T}Z} > \frac{K}{S(0)}\right].\end{aligned}$$

Since the natural logarithm is an increasing function, the above probability equals

$$\begin{aligned}\mathbb{P}^*\left[\left(r-\frac{\sigma^2}{2}\right)T+\sigma\sqrt{T}Z>\ln\left(\frac{K}{S(0)}\right)\right] &= \mathbb{P}^*\left[\sigma\sqrt{T}Z>\ln\left(\frac{K}{S(0)}\right)-\left(r-\frac{\sigma^2}{2}\right)T\right]\\ &= \mathbb{P}^*\left[Z>\frac{1}{\sigma\sqrt{T}}\left(\ln\left(\frac{K}{S(0)}\right)-\left(r-\frac{\sigma^2}{2}\right)T\right)\right]\\ &= \mathbb{P}^*\left[Z<-\frac{1}{\sigma\sqrt{T}}\left(\ln\left(\frac{K}{S(0)}\right)-\left(r-\frac{\sigma^2}{2}\right)T\right)\right]\\ &= \mathbb{P}^*\left[Z<\frac{1}{\sigma\sqrt{T}}\left(\ln\left(\frac{S(0)}{K}\right)+\left(r-\frac{\sigma^2}{2}\right)T\right)\right].\end{aligned}$$

We introduce the shorthand

$$d_2=\frac{1}{\sigma\sqrt{T}}\left(\ln\left(\frac{S(0)}{K}\right)+\left(r-\frac{\sigma^2}{2}\right)T\right),$$

which allows us to write our final result as

$$\mathbb{P}^*[S(T)>K]=N(d_2).$$

***Remark 18.6**: Of course, it's natural to ask the same question about the otherwise identical put. Here, we have*

$$\mathbb{P}^*[S(T)<K]=1-N(d_2)=N(-d_2).$$

Numerical Exercises.

Problem 18.7: *Assume the Black-Scholes model. The current stock price is* 50 *and the stock's volatility is* 0.25*. The stock pays no dividends. The continuously compounded risk-free interest rate is* 0.05*. Consider a European call option on this stock with strike price* 55 *and exercise date in six months.* Under the risk-neutral probability measure $\mathbb{P}^*$*, what is the probability that the call is* ***in-the-money*** *at expiration?*

Solution: *The call is in-the-money at expiration if* $S(T)>K$*. Under the risk-neutral probability measure,*

$$S(T)=S(0)e^{\left(r-\frac{1}{2}\sigma^2\right)T+\sigma\sqrt{T}Z}.$$

So, the required probability can be expressed as

$$\begin{aligned}\mathbb{P}^*[S(T)>K] &= \mathbb{P}^*\left[Z>\frac{1}{\sigma\sqrt{T}}\left(\ln\left(\frac{K}{S(0)}\right)-\left(r-\frac{1}{2}\sigma^2\right)T\right)\right]\\ &= \mathbb{P}^*[Z<d_2]=N(d_2),\end{aligned}$$

where

$$d_2 = \frac{1}{\sigma\sqrt{T}}\left[\ln\left(\frac{S(0)}{K}\right) + \left(r - \frac{1}{2}\sigma^2\right)T\right].$$

With our parameters,

$$d_2 = \frac{1}{0.25\sqrt{0.5}}\left[\ln\left(\frac{50}{55}\right) + \left(0.05 - \frac{1}{2}(0.25)^2\right)(0.5)\right] = \frac{-0.0953 + 0.0094}{0.1768} = -0.49.$$

So, our final answer is

$$\mathbb{P}^*[S(T) > 55] = N(-0.49) = 0.3121.$$

Problem 18.8: *Assume the Black-Scholes model. The current stock price is* 80 *and the stock's volatility is* 0.30*. The stock pays no dividends. The continuously compounded risk-free interest rate is* 0.05*. Consider a one-year,* 75*-strike European put option on this stock.* Under the risk-neutral probability measure $\mathbb{P}^*$*, what is the probability that the put is* ***in-the-money*** *at expiration?*

Solution: *The put is in-the-money at expiration if* $S(T) < K$*. Under the risk-neutral probability measure,*

$$S(T) = S(0)e^{(r-\frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z}.$$

So, the required probability can be expressed as

$$\begin{aligned}\mathbb{P}^*[S(T) < K] &= \mathbb{P}^*\left[Z < \frac{1}{\sigma\sqrt{T}}\left(\ln\left(\frac{K}{S(0)}\right) - \left(r - \frac{1}{2}\sigma^2\right)T\right)\right] \\ &= \mathbb{P}^*[Z < -d_2] = N(-d_2),\end{aligned}$$

where

$$d_2 = \frac{1}{\sigma\sqrt{T}}\left[\ln\left(\frac{S(0)}{K}\right) + \left(r - \frac{1}{2}\sigma^2\right)T\right].$$

With our parameters,

$$d_2 = \frac{1}{0.30}\left[\ln\left(\frac{80}{75}\right) + \left(0.05 - \frac{1}{2}(0.30)^2\right)\right] = \frac{0.0645 + 0.005}{0.30} = 0.23.$$

So, our final answer is

$$\mathbb{P}^*[S(1) < 75] = N(-0.23) = 0.4090.$$